

Datatype

September 8, 2026

1 TUPLE

Tuple is immutable

```
[2]: Tuple = (0,1,2,3,4,5)
```

```
[3]: Tuple[3]
```

```
[3]: 3
```

```
[4]: Tuple[:3]
```

```
[4]: (0, 1, 2)
```

```
[5]: Tuple[1:6:2]
```

```
[5]: (1, 3, 5)
```

```
[6]: Tuple[0] = 10
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[6], line 1  
----> 1 Tuple[0] = 10  
  
TypeError: 'tuple' object does not support item assignment
```

```
[7]: Tuple[5] + 5
```

```
[7]: 10
```

```
[8]: 5 in Tuple
```

```
[8]: True
```

```
[9]: 1 not in Tuple
```

```
[9]: False
```

```
[10]: Tuple2 = (5,6,7)
```

```
[11]: Tuple = (0,1,2,3,4,5)
      Tuple2 = (5,5,5,6,7)
      print(Tuple + Tuple2)
```

```
(0, 1, 2, 3, 4, 5, 5, 5, 5, 6, 7)
```

```
[ ]: len(Tuple2)
```

```
[ ]: 5
```

```
[ ]: Tuple2.index(5)
```

```
[ ]: 0
```

```
[ ]: Tuple2.count(5)
```

```
[ ]: 3
```

```
[15]: tuple1 = (10,) #single element tuple should end with comma
```

```
[16]: test = (1,[1,2])
```

```
[17]: test[1] = [1,2,3]
```

```
-----
TypeError                                Traceback (most recent call last)
Cell In[17], line 1
----> 1 test[1] = [1,2,3]

TypeError: 'tuple' object does not support item assignment
```

```
[18]: test[1][1] = [1,2,3]
```

```
[19]: test
```

```
[19]: (1, [1, [1, 2, 3]])
```

2 LIST

List is immutable

```
[44]: num1 = [1,2,3,4,5,6]
      str1 = ['a','b','c']
      mix1 = [1,2,6,45,'a','b'] # Mix datatype list
```

```
[45]: numl[2] is 10
```

```
<>:1: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?  
<>:1: SyntaxWarning: "is" with 'int' literal. Did you mean "=="?  
C:\Users\AV\AppData\Local\Temp\ipykernel_25276\4213587995.py:1: SyntaxWarning:  
"is" with 'int' literal. Did you mean "=="?  
    numl[2] is 10
```

```
[45]: False
```

```
[46]: strl[0] in mixl
```

```
[46]: True
```

```
[47]: strl[0] = 'b'  
      strl[1:] = ['c','d']
```

```
[48]: strl
```

```
[48]: ['b', 'c', 'd']
```

```
[49]: strl.insert(0,'a')
```

```
[50]: strl
```

```
[50]: ['a', 'b', 'c', 'd']
```

```
[51]: strl.append('e')
```

```
[52]: strl
```

```
[52]: ['a', 'b', 'c', 'd', 'e']
```

```
[53]: strl2 = ['f','g','h','i','i']
```

```
[54]: strl.extend(strl2)
```

```
[55]: strl
```

```
[55]: ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'i']
```

```
[56]: strl.remove('i')
```

```
[57]: strl
```

```
[57]: ['a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i']
```

```
[58]: strl.pop(2)
```

```
[58]: 'c'
```

```
[59]: str1
```

```
[59]: ['a', 'b', 'd', 'e', 'f', 'g', 'h', 'i']
```

```
[ ]: str1.pop()
```

```
[ ]: 'i'
```

```
[ ]: str1
```

```
[ ]: ['a', 'b', 'd', 'e', 'f', 'g', 'h']
```

```
[62]: del str12[2]
```

```
[ ]: str12
```

```
[ ]: ['f', 'g', 'i', 'i']
```

```
[64]: str12.clear()
```

```
[65]: str12
```

```
[65]: []
```

```
[66]: del str12
```

```
[67]: str12
```

```
-----  
NameError                                Traceback (most recent call last)  
Cell In[67], line 1  
----> 1 str12  
  
NameError: name 'str12' is not defined
```

2.1 Challenge

Even no from list

```
[68]: n01 = [1,2,3,4,5,6,7,8,9,10]
```

```
[69]: for i in n01:  
        if i % 2 == 0:  
            print(i, end=" ")
```

2 4 6 8 10

```
[70]: # List Comprehension
even = [x for x in nol if x % 2 == 0]
print (even)
```

[2, 4, 6, 8, 10]

Listt Comprehention

```
[ 'var' for 'var' in 'LIST' 'Condition' ]
```

2.2 Challenge

Leap Year from list

```
[71]: year = [2000,1900,2024,2004,2026]
```

```
[72]: for i in year:
        if i % 100 ==0:
            if i % 400 == 0:
                print(i, "is a leap year")
            else:
                print(i, "is a not leap year")
        else:
            if i % 4 == 0:
                print(i, "is a leap year")
            else:
                print(i, "is a not leap year")
```

2000 is a leap year
1900 is a not leap year
2024 is a leap year
2004 is a leap year
2026 is a not leap year

```
[ ]:
```